

# WINDOWS: VISUAL STUDIO AND QT CREATOR

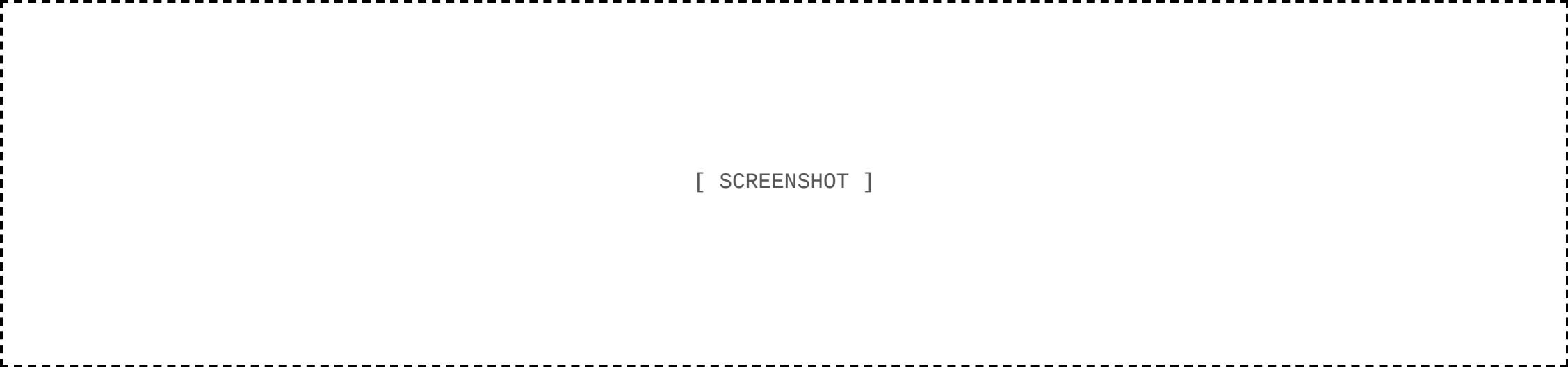
Two mainstream options, one compiler in common

## Your choice on Windows

- **Visual Studio** - free Community edition, ships its own compiler (MSVC).
- **Qt Creator** - a lighter, cross-platform IDE. On Windows it doesn't ship a compiler of its own - it needs you to pick a **Kit**.

Both read the same `CMakeLists.txt` from this course. Pick whichever one you're more comfortable with.

## Installing Visual Studio

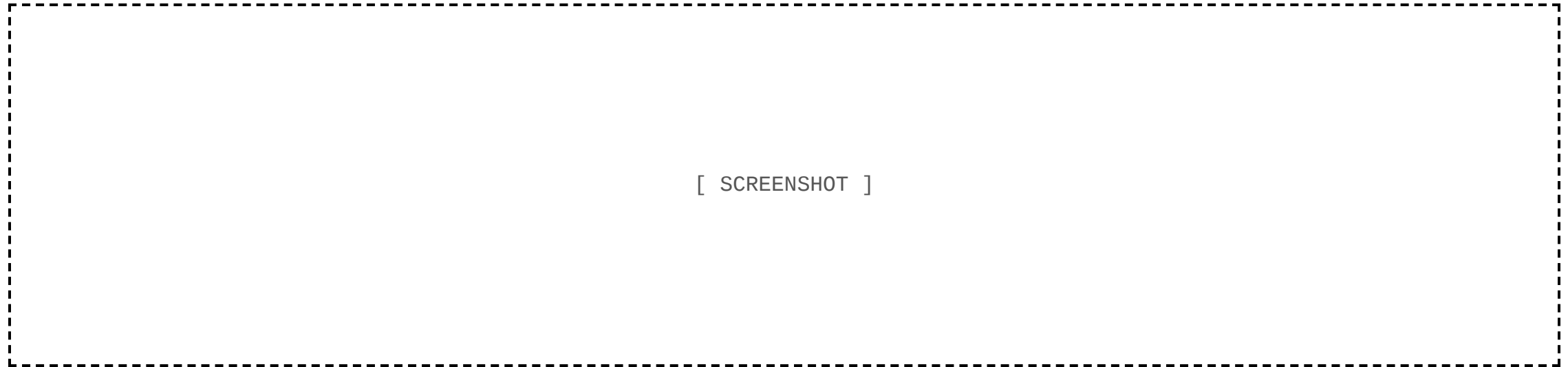


[ SCREENSHOT ]

1. Download the Visual Studio Installer, choose **Community** (free).
2. Under **Workloads**, check **Desktop development with C++**.
3. That workload includes MSVC, the Windows SDK, and CMake support - you don't install those separately.

## Open any lecture folder directly

Visual Studio can open a folder containing a `CMakeLists.txt` without a `.sln` file at all: **File -> Open -> CMake...** and pick this lecture's folder.



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Visual Studio detects `CMakeLists.txt`, configures the project, and shows `rooster` as the run/debug target.

MSVC

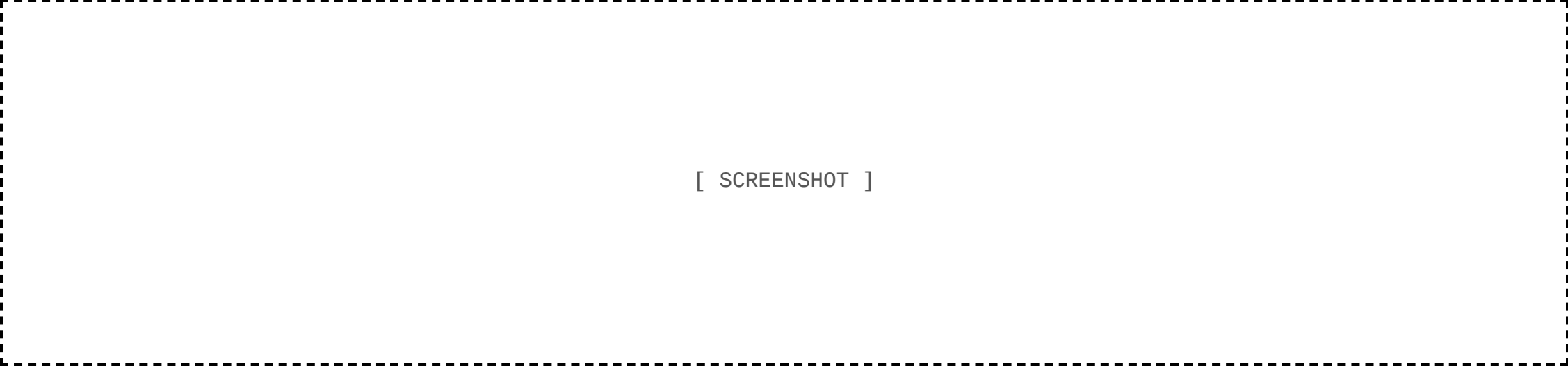
## MSVC - the compiler that comes with it

Visual Studio's C++ compiler is called **MSVC**. Its C++23 support is controlled by a compiler flag, not a version number you pick yourself:

```
/std:c++latest
```

CMake's `set(CMAKE_CXX_STANDARD 23)` (already in this course's `CMakeLists.txt`) asks CMake to add the right flag for you automatically.

## Installing Qt Creator



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The Qt Online Installer lets you select **Qt Creator** as a standalone component - you don't need to install the full Qt framework or any Qt libraries just to use it as a C++ IDE.

## KITS

### Kits - a bundle Qt Creator needs to build anything

A **Kit** = a compiler + a debugger + (optionally) a Qt version, bundled together under one name. Qt Creator can't build a project until at least one Kit is configured.

On Windows, the installer usually detects one automatically if Visual Studio is already installed (an **MSVC kit**). You can also add:

- a **MinGW** kit (a GCC build for Windows)
- an **LLVM/Clang** kit, if you've installed one separately

## Opening this lecture in Qt Creator

[ SCREENSHOT ]

**File -> Open File or Project**, choose this folder's **CMakeLists.txt**.

Qt Creator shows a **Configure Project** screen listing every Kit it knows about - check the one you want to build with, then **Configure Project**.

DIFFERENT KITS, DIFFERENT SUPPORT

## **Not all kits support C++23 equally**

MSVC, MinGW, and LLVM/Clang kits are three genuinely different compilers - each with its own pace of adding new C++ features. A kit that's a version or two behind may not support everything this course uses yet.

Lecture 2.5 shows how to check, for whichever kit you end up using.